

Quantum Electrodynamics of Bound States in Strong Coulomb Fields (Page 1): Feynman diagram for one-loop electron self-energy

J. R. Oppenheimer and H. A. Bethe - Institute for Advanced Study

Section 3: Radiative Shift Calculations

Abstract

The Lamb shift energy difference ΔE_L is given by:

$$\Delta E_L = \frac{4\pi e^2}{3} \frac{1}{n^3} \int_0^\infty \frac{F(Z\alpha)}{k} dk$$

We calculate radiative corrections to electron binding energies in hydrogenic ions.

Section 1: Introduction and Field Equations

The Dirac Hamiltonian in an external four-vector potential A_μ is defined as

$$H_D = \alpha \cdot (\mathbf{p} - e \mathbf{A}) + \beta m + e A_0$$

The relativistic S-matrix expansion yields the self-energy operator $\Sigma(p)$

$$\Sigma(p) = -i e^2 \int \frac{d^4 k}{(2\pi)^4} \gamma^\mu S_F(p-k) \gamma_\mu D_F(k)$$

Section 2: Vacuum Polarization Tensor

The regularized polarization tensor $\Pi^{\mu\nu}(q)$ satisfies current conservation

$$q_\mu \Pi^{\mu\nu}(q) = 0$$

Table 1: Theoretical vs experimental binding energies in eV for Z=1

State	Dirac Energy (eV)	Self Energy (eV)	Net Bound Energy (eV)
1s1/2	-132279.9	+355.1	-131924.8
2s1/2	-35122.09	+58.3	-35063.79

Figure 2 (Page 4): Comparison of calculated self-energy shifts across different states

Section 4: Concluding Remarks and Higher Order Perturbation

Higher order corrections remain perturbative even in the critical coupling regime